

NON-STATIONARY APPROACH - NUMERICAL SOLUTION

Complete Step-by-Step Walkthrough

PROBLEM SCENARIO

Setup

- **5-armed bandit** with 20 total plays
 - **Initial Distribution (Plays 1-10):**
 - Arm 0: Probability 0.30 → Average reward 3.0
 - Arm 1: Probability 0.50 → Average reward 5.0
 - Arm 2: Probability 0.60 → Average reward 6.0 **BEST IN PHASE 1**
 - Arm 3: Probability 0.20 → Average reward 2.0
 - Arm 4: Probability 0.45 → Average reward 4.5
 - **Shifted Distribution (Plays 11-20) - ENVIRONMENT CHANGES!:**
 - Arm 0: Probability 0.60 → Average reward 6.0 **NOW BEST!**
 - Arm 1: Probability 0.40 → Average reward 4.0 (decreased)
 - Arm 2: Probability 0.50 → Average reward 5.0 (decreased)
 - Arm 3: Probability 0.30 → Average reward 3.0 (improved)
 - Arm 4: Probability 0.55 → Average reward 5.5
 - **Strategy:** -Greedy ($\epsilon = 0.2$) with **constant learning rate** $= 0.1$
 - **Key:** Instead of $1/N$ (decaying), use fixed ϵ for recency bias
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WHY NON-STATIONARY?

Traditional Approach ($1/N$ step size)

Step at play t : $1/N(a) \rightarrow$ decreases to 0

Problem in changing environment:

- Old data (plays 1-5) weighted equally with new data (plays 16-20)
- Can't adapt when distributions shift
- Gets stuck in old beliefs

Non-Stationary Approach (constant ϵ)

Step at every play: $\epsilon = 0.1 \rightarrow$ constant!

Benefit in changing environment:

- Recent data gets more weight

- Older data "fades away" exponentially
- Can adapt when best arm changes

LEARNING RATE ANALYSIS

Memory Decay with $\alpha = 0.1$

How much weight does each past reward have?

Current effect from:

Current reward:	$100\% \times 0.1 = 10\%$
1 play ago:	$90\% \times 10\% = 9\%$
2 plays ago:	$81\% \times 10\% = 8.1\%$
5 plays ago:	$59\% \times 10\% = 5.9\%$
10 plays ago:	$35\% \times 10\% = 3.5\%$
20 plays ago:	$12\% \times 10\% = 1.2\%$

Effective memory $1/\alpha = 1/0.1 = 10$ plays

(Recent data matters much more than old data)

Comparison: $\alpha = 0.1$ vs $1/N$

After 10 plays: - $1/N$ step size: $1/10 = 0.1$ (same as $\alpha = 0.1$ at this point)

After 20 plays: - $1/N$ step size: $1/20 = 0.05$ (gets smaller!) - $\alpha = 0.1$ step size: 0.1 (stays constant!)

This difference becomes critical when environment shifts at play 11!

COMPLETE NUMERICAL WALKTHROUGH

PHASE 1: PLAYS 1-10 (Original Distribution)

PLAY 1 Selection: Arm 0 (via ϵ -greedy, $\epsilon = 0.2$, $r = 0.75 \rightarrow$ exploit) **Reward:** Arm 0 with $p=0.30$, draw=0.42 $\rightarrow$ **Reward: 0**

Non-stationary update:

$Q_{old}(0) = 0.0$

$error = 0 - 0.0 = 0$

$Q_{new}(0) = 0.0 + 0.1 \times 0 = 0.0$

State: $Q = [0.0, 0.0, 0.0, 0.0, 0.0]$, $N = [1, 0, 0, 0, 0]$

PLAY 2 Selection: Arm 3 (via random explore, $r = 0.08 < 0.2$) **Reward:** Arm 3 with $p=0.20$, draw=0.55 $\rightarrow$ **Reward: 0**

Non-stationary update:

$Q_{\text{old}}(3) = 0.0$
 $\text{error} = 0 - 0.0 = 0$
 $Q_{\text{new}}(3) = 0.0 + 0.1 \times 0 = 0.0$

State: $Q = [0.0, 0.0, 0.0, 0.0, 0.0]$, $N = [1, 0, 0, 1, 0]$

PLAY 3 Selection: Arm 1 (via exploit) **Reward:** Arm 1 with $p=0.50$, draw=0.38 $\rightarrow$ **Reward: 10**

Non-stationary update:

$Q_{\text{old}}(1) = 0.0$
 $\text{error} = 10 - 0.0 = 10$
 $Q_{\text{new}}(1) = 0.0 + 0.1 \times 10 = 1.0 \leftarrow \text{IMPORTANT: Only gets } 0.1 \times \text{reward!}$

State: $Q = [0.0, 1.0, 0.0, 0.0, 0.0]$ **Note:** With $\alpha=0.1$, we're conservative (remember: $1/N$ would give 10.0)

PLAY 4 Selection: Arm 1 (via exploit, highest Q) **Reward:** Arm 1 with $p=0.50$, draw=0.61 $\rightarrow$ **Reward: 0**

Non-stationary update:

$Q_{\text{old}}(1) = 1.0$
 $\text{error} = 0 - 1.0 = -1.0$
 $Q_{\text{new}}(1) = 1.0 + 0.1 \times (-1.0) = 1.0 - 0.1 = 0.9$

State: $Q = [0.0, 0.9, 0.0, 0.0, 0.0]$

PLAY 5 Selection: Arm 2 (via random explore) **Reward:** Arm 2 with $p=0.60$, draw=0.35 $\rightarrow$ **Reward: 10**

Non-stationary update:

$Q_{\text{old}}(2) = 0.0$
 $\text{error} = 10 - 0.0 = 10$
 $Q_{\text{new}}(2) = 0.0 + 0.1 \times 10 = 1.0$

State: $Q = [0.0, 0.9, 1.0, 0.0, 0.0]$ **Critical:** Found Arm 2 (best in phase 1)

PLAYS 6-10 SUMMARY Continuing with similar patterns:

Play	Arm	Reward	Error	Q Before	Q After
6	2	10	0.1	10-1.0	1.0
7	0	10	0.1	10-0	0.0
8	2	10	0.1	10-1.9	1.9
9	4	0	0.1	0-0	0.0
10	2	0	0.1	0-2.71	2.71

End of Phase 1:

Q = [1.0, 0.9, 2.44, 0.0, 0.0]

Total Reward Phase 1: ~30

Note: Q-values are LOW (conservative =0.1) but Arm 2 is clearly highest

CRITICAL MOMENT: ENVIRONMENT SHIFT AT PLAY 11

What agent believes (old environment):

Q = [1.0, 0.9, 2.44, 0.0, 0.0]

Arm 2 looks best!

What really happened (new environment starts):

Arm 0: Now probability 0.60 (was 0.30)

Arm 2: Now probability 0.50 (was 0.60)

Arm 0 is NOW the best!

But agent doesn't know this yet...

PHASE 2: PLAYS 11-20 (Shifted Distribution)

PLAY 11 Selection: Arm 2 (via exploit, highest Q=2.44) **Reward (NEW ENVIRONMENT):** Arm 2 with p=0.50, draw=0.42 → **Reward: 10**

Non-stationary update:

Q_old(2) = 2.44

error = 10 - 2.44 = 7.56 (LARGE positive surprise!)

Q_new(2) = 2.44 + 0.1 × 7.56 = 2.44 + 0.756 = 3.196

But wait - Arm 2's TRUE mean just dropped from 6.0 to 5.0!

Got lucky with this pull.

State: Q = [1.0, 0.9, 3.196, 0.0, 0.0] **Total Reward Phase 2:** 10

PLAY 12 Selection: Arm 2 (still highest Q) **Reward (NEW ENV):** Arm 2 with $p=0.50$, draw=0.72 $\rightarrow$ **Reward: 0**

Non-stationary update:

$$Q_{\text{old}}(2) = 3.196$$

$$\text{error} = 0 - 3.196 = -3.196$$

$$Q_{\text{new}}(2) = 3.196 + 0.1 \times (-3.196) = 3.196 - 0.320 = 2.876$$

State: $Q = [1.0, 0.9, 2.876, 0.0, 0.0]$ **Q(2) dropping! Adaptation starting!**

PLAY 13 Selection: Arm 0 (via random explore, $r < 0.2$) **Reward (NEW ENV):** Arm 0 with $p=0.60$, draw=0.45 $\rightarrow$ **Reward: 10**

Non-stationary update:

$$Q_{\text{old}}(0) = 1.0$$

$$\text{error} = 10 - 1.0 = 9.0 \text{ (Big surprise!)}$$

$$Q_{\text{new}}(0) = 1.0 + 0.1 \times 9.0 = 1.0 + 0.9 = 1.9$$

DISCOVERY! Arm 0 is looking better!

State: $Q = [1.9, 0.9, 2.876, 0.0, 0.0]$ **Critical:** Exploration found the new best arm!

PLAY 14 Selection: Arm 2 (still highest $Q=2.876$) **Reward (NEW ENV):** Arm 2 with $p=0.50$, draw=0.68 $\rightarrow$ **Reward: 0**

Non-stationary update:

$$Q_{\text{old}}(2) = 2.876$$

$$\text{error} = 0 - 2.876 = -2.876$$

$$Q_{\text{new}}(2) = 2.876 + 0.1 \times (-2.876) = 2.588$$

State: $Q = [1.9, 0.9, 2.588, 0.0, 0.0]$ **Arm 2 continuing to decline**

PLAY 15 Selection: Arm 0 (via exploit, now highest $Q=1.9$) **Reward (NEW ENV):** Arm 0 with $p=0.60$, draw=0.25 $\rightarrow$ **Reward: 10**

Non-stationary update:

$$Q_{\text{old}}(0) = 1.9$$

$$\text{error} = 10 - 1.9 = 8.1$$

$$Q_{\text{new}}(0) = 1.9 + 0.1 \times 8.1 = 1.9 + 0.81 = 2.71$$

Arm 0 rising! Switching preference!

State: $Q = [2.71, 0.9, 2.588, 0.0, 0.0]$ **Arm 0 now higher than Arm 2!**

PLAYS 16-20 SUMMARY Agent continues adapting to new environment:

Play	Arm	Reward	Error	Q Before	Q After	Note
16	0	10	8.29	2.71	3.54	Rising
17	0	0	-3.54	3.54	3.18	Variance
18	0	10	6.82	3.18	3.87	Rising
19	0	10	6.13	3.87	4.48	Rising
20	2	10	7.52	2.588	3.34	Late luck

End of Phase 2:

$Q = [4.48, 0.9, 3.34, 0.0, 0.0]$

Total Reward Phase 2: ~60

Total Reward Overall: ~90

ADAPTATION ANALYSIS

How Non-Stationary Adapted

Phase 1 (Original): Arm 2 best ($Q=2.44$) **Shift at Play 11:** Environment changed **Phase 2 (New):** Arm 0 became best

Q-Value Trajectories:

Arm 0: $0.0 \rightarrow 1.0 \rightarrow 1.9 \rightarrow 2.71 \rightarrow 3.54 \rightarrow 4.48$ Rising

Arm 2: $0.0 \rightarrow 1.0 \rightarrow 2.44 \rightarrow 3.196 \rightarrow 2.876 \rightarrow 2.588 \rightarrow 3.34$ Declining then stabilizing

Recency Bias in Action

By play 20, Arm 0's Q reflects: - Mostly recent plays (11-20) with new $p=0.60$ - Less weight on old plays (1-10) from old $p=0.30$ - Adapted successfully to environment shift!

Arm 2's Q reflects: - Mixture of old (5.0 true mean in phase 2) and new (6.0 old phase) - But declining trend visible - Could eventually realize it's not best if continued

COMPARISON: CONSTANT VS $1/N$

Same rewards, different step sizes:

With $1/N$ (stationary approach):

Play 11: Step size = $1/N(2)$ (depends on how many times tried)

Play 20: Step size = $1/N(2)$ (gets smaller, not adaptive)

Problem: Old data (phase 1) still weighted in average!

With $\alpha = 0.1$ (non-stationary):

Play 11: Step size = 0.1 (constant, responsive)

Play 20: Step size = 0.1 (still responsive to changes)

Benefit: Recent data dominates, can adapt!

Long-term Regret

With $1/N$ (locked in old understanding):

Even at play 50:

- Q-value still weighted toward phase 1 means
- Takes hundreds of plays to forget old data
- Regret keeps growing

With $\alpha = 0.1$ (adapts quickly):

At play 20:

- Already strongly adapting to new environment
- Will exploit new best arm soon
- Lower regret potential

FINAL PERFORMANCE METRICS

Results Over 20 Plays

Metric	Value
Total Reward Phase 1	~30
Total Reward Phase 2	~60
Total Reward Overall	~90
Arms Explored	5/5 (via ϵ -greedy)
Best Arm Phase 1 Found	Arm 2
Best Arm Phase 2 Found	Arm 0
Adaptation Speed	Fast (by play 15)

Comparison with Stationary (1/N)

Stationary would: - Get stuck on Arm 2 (best in phase 1) - Slowly realize it's not best anymore - Take many more plays to switch - Higher regret (~70 vs 90)

Non-stationary does: - Discovers Arm 2 works well initially - Through exploration, finds Arm 0 - Quickly switches preference - Adapts within phase 2 itself - Better long-term (~90 total)

NUMERICAL FORMULAS USED

Non-Stationary Q-Update

$$Q_{new}(a) = Q_{old}(a) + \alpha \times (R_t - Q_{old}(a))$$

Where: $\alpha = 0.1$ (constant learning rate) - Error = $R_t - Q_{old}(a)$ - Step size = α (not $1/N!$)

Recency Weight Formula

Weight of reward from t plays ago:
 $weight = \alpha \times (1 - \alpha)^t$

With $\alpha = 0.1$:

Recent ($t=0$):	0.1
$t=1$:	$0.1 \times 0.9 = 0.09$
$t=5$:	$0.1 \times 0.9^5 = 0.059$
$t=10$:	$0.1 \times 0.9^{10} = 0.035$

KEY INSIGHTS

Non-Stationary Advantages

1. **Adapts to shifts:** Doesn't get locked in old beliefs
2. **Recency bias:** Recent data weighted heavily
3. **Responsive:** Immediate reaction to changes
4. **Tunable:** α parameter controls adaptation speed

Parameter Selection for

$\alpha = 0.01$: Very conservative (remember ~100 plays)
 $\alpha = 0.05$: Conservative (remember ~20 plays)
 $\alpha = 0.10$: Moderate ← Used here
 $\alpha = 0.20$: Aggressive (remember ~5 plays)
 $\alpha = 0.50$: Very aggressive (remember ~2 plays)

Rule: Higher α = more adaptive to changes = less memory

When to Use Non-Stationary

- Market dynamics (prices, trends change)
 - Seasonal effects (demand varies by season)
 - User behavior evolution (preferences change)
 - Competitor adaptation (rivals change strategies)
 - Any domain with concept drift
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SUMMARY

What Happened

1. **Phase 1:** Environment stable, found Arm 2 as best
2. **Play 11: Environment shifted** (Arm 0 now best)
3. **Plays 12-13:** Started adapting ($Q(2)$ down, $Q(0)$ up)
4. **Plays 14-20:** Continued adaptation, switched to Arm 0
5. **Phase 2 result:** Successfully found new best arm

Why Non-Stationary Worked

Constant $\alpha = 0.1$ gave **recency bias**: - Old plays (phase 1) faded exponentially
- New plays (phase 2) had full impact - Could adapt when environment changed
- Didn't get locked into old beliefs

Lesson

In dynamic environments, constant step size (α) beats decaying step size ($1/N$). The ability to forget old data and adapt to new realities is essential for survival when the world changes.

This is why non-stationary approaches are critical for real-world applications like stock trading, online advertising, and recommendation systems!